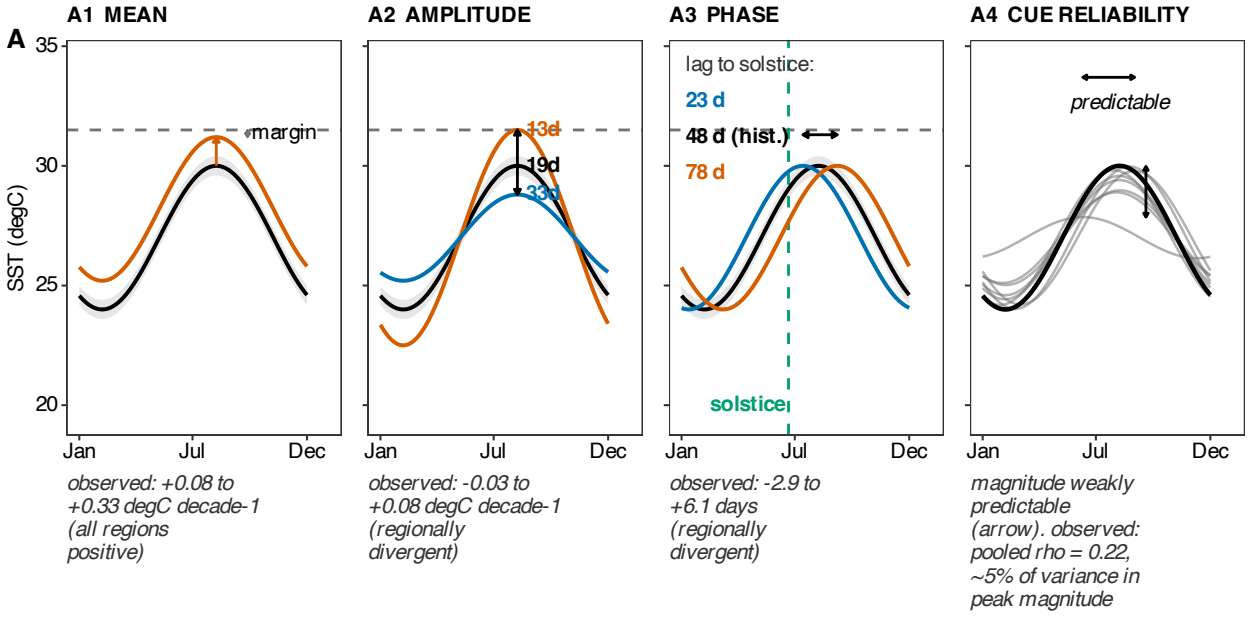
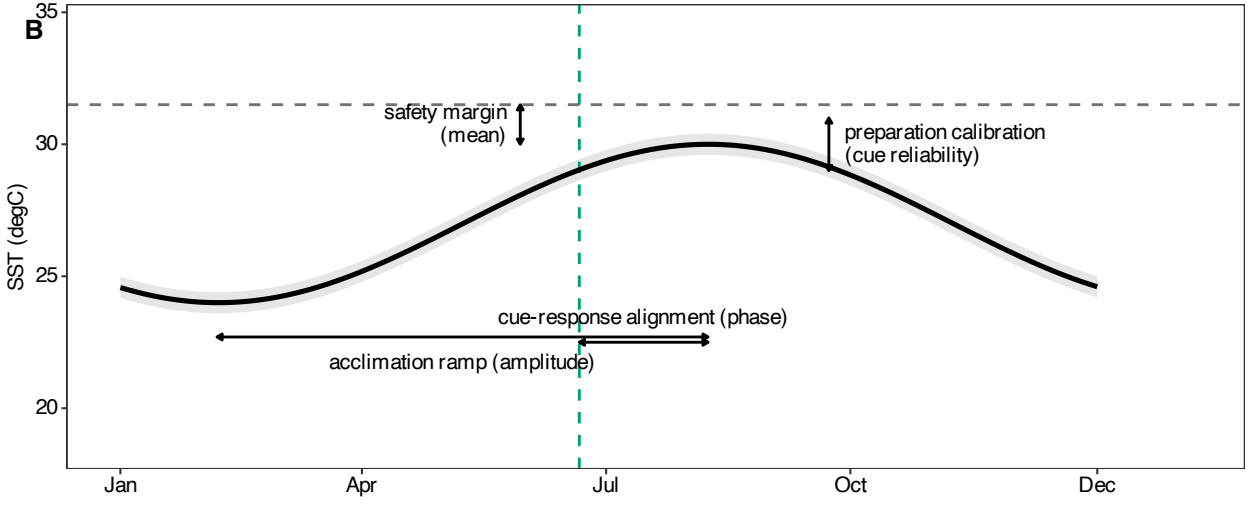
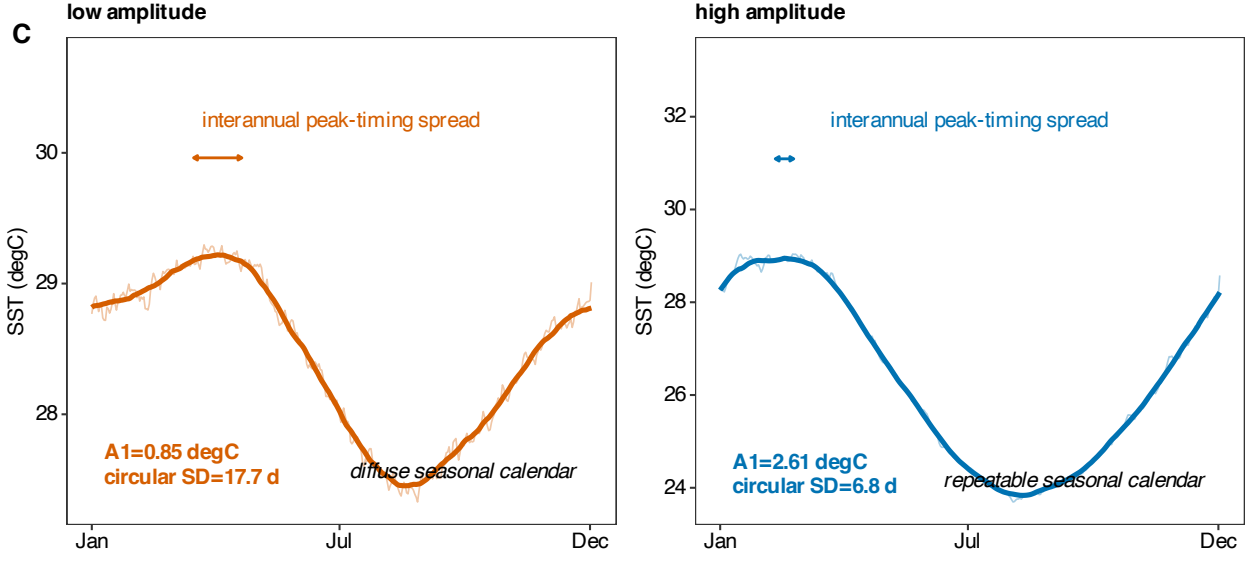

A
A1 MEAN
A2 AMPLITUDE
A3 PHASE
A4 CUE RELIABILITY
SST (degC)
35
30
25
20
margin
13d
19d
33d
lag to solstice:
23 d
48 d (hist.)
78 d
solstice
predictable
Jan
Jul
Dec
observed: +0.08 to +0.33 degC decade-1 (all regions positive)
observed: -0.03 to +0.08 degC decade-1 (regionally divergent)
observed: -2.9 to +6.1 days (regionally divergent)
magnitude weakly predictable (arrow). observed: pooled rho = 0.22, ~5% of variance in peak magnitude


B
SST (degC)
35
30
25
20
safety margin (mean)
preparation calibration (cue reliability)
cue-response alignment (phase)
acclimation ramp (amplitude)
Jan
Apr
Jul
Oct
Dec


C
low amplitude
high amplitude
SST (degC)
interannual peak-timing spread
A1=0.85 degC
circular SD=17.7 d
diffuse seasonal calendar
A1=2.61 degC
circular SD=6.8 d
repeatable seasonal calendar
Jan
Jul
Dec